Numerical assignement 2 | Intro to CMP | 2026
Nearly free electron model

Redo the midsem question 4 numerically !
Let $V(x) = A\cos(\pi x/a) + (A/2)\cos(2\pi x/a) + (A/3)\cos(3\pi x/a)$ with A=3eV.

1. Plot the band structure in the first and second BZ for at least 5 bands.
2. Plot the corresponding density of states.
3. Converge the band-structure w.r.t the increasing number of Fourier basis used for expansion of the Bloch states. (In class we mostly wrote it for five plane wave basis e^{iGx} with G from G_2 to G_{-2}.
4. Mark the Fermi energy if each unitcell has one valence electrons.
   Converge the Fermi energy w.r.t the increasing number unit cell considered in the supercell, or in other words, finer k mesh.
5. Next add the potential $2A\cos(0.5\pi x/a)$. Plot bandstructure and density of states.
6. Compute total energy per unit cell by simply adding the energy of the occupied states and dividing it by N_{BVK}. Plot the energy per unitcell as a function of A.
   Note however that this is of course the most rudimentory and approximate approach to compute total energy since we are not considering any interaction among the electrons.

One simple approach to improve on the computation of electronic structure within the single particle picture is to solve the Poisson equation to obtain the potential due to the charge density of the occupied states and add it to V(x) and iteratively converge the charge density.
You can easily do it by writing the Poisson equation in the Fourier basis.
You have a break coming up so go for it if you are intersted !